

Nimay Vyas

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SENIOR DATA/SOFTWARE ENGINEER | DISTRIBUTED SYSTEMS SPECIALIST | AI DATA PLATFORM ENGINEER

PRODUCT OWNER PROFILE

Product-focused technology professional with experience owning product strategy and roadmap priorities, translating customer and business needs into requirements, collaborating with Agile engineering teams, and delivering data, API, and distributed-system capabilities. Experienced with REST APIs, GraphQL, microservices, access-control scenarios, data governance, Jira, and cross-functional stakeholder collaboration.

EDUCATION

University of Maryland, College Park

Bachelor of Science in Computer Science

College Park, MD

Aug. 2024

Academic Honors: Northrop Grumman Scholarship, OMSE Academic Excellence Award, Dean's List

PROFESSIONAL EXPERIENCE

Data Engineer

Aug. 2025 - Present

Ernst & Young

Newark, NJ

- Designed and operated production Python and SQL reconciliation and ETL pipelines processing 7,000,000+ fixed-income records, supporting enterprise reporting, analytics, and downstream data consumers.
- Redesigned relational schemas and optimized SQL reconciliation logic, improving query performance by 70% while strengthening data integrity across downstream workflows.
- Built Gold-layer pipelines for Security Dim, Issuer Dim, and Coupon Current Dim, standardizing multi-source reference data through reusable transformations, historical backfills, and automated validation.
- Developed ETL/ELT pipelines in Scala and Spark to ingest, transform, validate, and process structured and semi-structured data.
- Developed and deployed machine learning and deep learning models using TensorFlow for classification, regression, and predictive analytics.
- Modernized legacy SQL by converting Sybase-specific logic to ANSI-compliant patterns and automated production scheduling with Autosys, improving portability and reducing manual intervention.

Data Engineer - Aladdin Data

Aug. 2024 - Aug. 2025

BlackRock

Wilmington, DE

- Developed and optimized Kafka-based distributed data pipelines Python for real-time financial data workflows, improving processing throughput by 200% across Aladdin data services.
- Implemented advanced SQL logic and backend enhancements within the QDFS dataflow server, improving data accessibility, query reliability, and large-scale processing performance.
- Documented RLS/CLS test scenarios, expected results, defects, and evidence to support **data governance and audit requirements**.
- Designed and implemented scalable **AWS data pipelines** using **Amazon S3, AWS Glue, Athena, Kinesis, EventBridge, and PySpark** for batch and near-real-time data processing.
- Integrated Scala-based Spark applications with **Kafka, AWS S3, Hive, and relational databases** for real-time and batch data processing.
- Built exception-handling and reconciliation processes that routed data-quality breaks to operators, reduced recurring production issues, and improved operational visibility.
- Validated **positive and negative access scenarios** for RLS/CLS, including authorized, unauthorized, privileged, and cross-role users.
- Connected Tableau to SQL-based data sources and optimized analytical datasets to improve dashboard performance and refresh reliability for internal platform users.

Founder & CEO

July. 2020 - Aug. 2024

NitesOut

College Park, MD

- Founded and scaled a South Asian curated experiences startup supported by UMD Startup Shell and the Dingman Center for Entrepreneurship, generating \$200,000+ in revenue and attracting 10,000+ attendees.
- Owned company strategy, event economics, partnerships, marketing, and cross-functional operations, with accountability for revenue growth, attendance, and end-to-end execution.
- Built scalable **ETL/ELT pipelines** using Spark DataFrames, Spark SQL, RDDs, and Spark Streaming.
- Implemented data transformations, joins, aggregations, filtering, and partitioning to process large datasets efficiently.
- Developed reusable and modular Scala components following **object-oriented and functional programming principles**.
- Guided a product team in building and launching a viable web and mobile MVP with React, React Native, and Tailwind CSS, translating customer and operator needs into product requirements and roadmap priorities.

Data Engineer Intern

June 2023 - Aug. 2023

JPMorgan Chase

Wilmington, DE

- Built backend microservices and workflow automation features using Java, Spring Boot, Apache Camel, Kafka, and React.js for enterprise skill-management operations.
- Implemented Kafka-based event-driven processing for asynchronous change requests, strengthening scalability, reliability, and communication across distributed services.
- Collaborated with data scientists and engineers to integrate TensorFlow-based models into **end-to-end AI/ML applications**.
- Developed reusable Spark frameworks and libraries to standardize data processing across multiple projects.
- Partnered with Agile engineering teams to design, test, and deliver a micro-application that reduced manual back-office workflow steps for internal business users.
- Integrated Apache Spark with Kafka, Hadoop, Hive, Snowflake, and other data engineering technologies to support enterprise data platforms.

Data Engineer

U.S. Food & Drug Administration

Jan 2022 - Jun. 2023

Silver Spring, MD

- Curated, analyzed, and validated large clinical and testing datasets for approved ophthalmic devices, improving data quality and reporting accuracy.
- Applied statistical analysis and structured data checks to support regulatory reporting and evidence-based decision-making.
- Implemented **Spark SQL** queries for complex data transformations, aggregations, joins, and analytical workloads.
- Used Scala collections, **case classes**, **traits**, **pattern matching**, **higher-order functions**, and **Futures** to build robust applications.

Data Engineer

Maxim Healthcare

Jan 2022 - Jan. 2022

Columbia, MD

- Developed optimized SQL queries and SSRS reports in SSMS, enabling business intelligence, operational reporting, and data-driven decision-making for enterprise users.

SELECTED PRODUCTION & PRODUCT INITIATIVES

Gold-Layer Security Data Pipelines / Python, SQL, ETL, Data Modeling, Autosys

Ernst & Young

- Engineered production pipelines for Security Dim, Issuer Dim, and Coupon Current Dim, ingesting multiple upstream files, applying standardized transformations, and publishing governed reference data for downstream reporting.
- Added historical backfill and reconciliation controls to compare source-to-target results, isolate breaks, and support repeatable data-quality analysis.

Aladdin Exception Handling & Reconciliation Framework / Python, SQL, Kafka, Distributed Systems

BlackRock

- Designed exception routing, operator alerts, and reconciliation logic for Aladdin data workflows, improving detection and resolution of recurring production data-quality issues.
- Enhanced backend dataflow services and SQL processing to support scalable financial data analysis across platform consumers.

SELECTED PRODUCTION & PRODUCT INITIATIVES - CONTINUED

Cloud-Native ELT & Lakehouse Modernization / AWS Glue, PySpark, S3, Databricks, Delta Lake, Snowflake, dbt

Ernst & Young

- Designed cloud-native ingestion and transformation patterns using Parquet-based storage, reusable SQL models, and curated data layers for analytics and machine-learning consumers.
- Developed reusable and modular Scala components following **object-oriented and functional programming principles**.
- Integrated data-lake and warehouse components to improve scalability, secure data sharing, and repeatability across high-volume financial workflows.
- Worked with AWS S3, Azure Data Lake, HDFS, Hive, and cloud-based data platforms for scalable data processing.

Claims Data Extraction & Analytics Enablement / SQL, Tableau, ETL, Data Validation

Ernst & Young

- Extracted and transformed large-scale claims data from relational databases, then structured datasets for downstream reporting, dashboarding, and analytical review.
- Applied query optimization and validation checks to improve refresh reliability and traceability across reporting workflows.

NitesOut Web & Mobile MVP / React, React Native, Tailwind CSS, Product Strategy

NitesOut

- Led MVP definition and delivery across web and mobile experiences, aligning product requirements with event discovery, ticketing, customer engagement, and internal operating needs.
- Prioritized features through customer feedback and business impact, creating a viable foundation for a scalable curated-experiences platform.

QCaaS Exceptions Analytics Dashboard / Python, Streamlit, Pandas, SQL, Altair

BlackRock

- Built a multi-tab operational dashboard with client and exception-subtype filters, variance analysis, status distributions, and historical exception counts across recent processing runs.
- Added exception logging and client-level drilldowns to help operators isolate data breaks, compare source and quality-control outputs, and prioritize remediation.

Event-Driven Data Processing Architecture / Kafka, Kinesis, EventBridge, Firehose, Microservices

JP Morgan Chase

- Designed streaming and asynchronous processing patterns for near-real-time transaction and investment-data ingestion across distributed services.
- Improved system resilience by decoupling producers and consumers, supporting scalable workflow orchestration and fault-isolated processing.

Master Data Management & Golden Source / SQL, Data Modeling, Snowflake, Governance

Ernst & Young

- Supported Golden Source data models for security, portfolio, issuer, and reference data, improving consistency across downstream financial applications.
- Implemented reconciliation and governance controls to standardize definitions, validate transformations, and strengthen source-to-target traceability.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, scala, C++, C, Rust, Ruby, HTML, CSS

Data Engineering: Apache Kafka, ETL/ELT, Data Pipelines, Data Modeling, Batch Processing, Streaming, Reconciliation, Data Quality, Data Governance, Autosys, PySpark, dbt

Cloud & Data Platforms: AWS, Amazon S3, AWS Glue, Athena, Kinesis, EventBridge, Azure, Snowflake, TensorFlow, Databricks, Delta Lake, Docker, Kubernetes, Terraform

Frameworks & APIs: REST APIs, GraphQL, Microservices, Spring Boot, Apache Camel, Node.js, Express.js, React.js, React Native, Tailwind CSS

AI & Analytics: Generative AI, LLMs, RAG, NLP, PyTorch, scikit-learn, Pandas, NumPy, Tableau, SSRS

Databases & Tools: Jira, Confluence, Microsoft SQL Server, SSMS, MongoDB, Relational/NoSQL Databases, Git, Bitbucket, Jenkins, Maven, CI/CD, Linux/Unix, JUnit, Mockito

Business & Product: Product Vision, Product Roadmaps, Requirements Gathering, Feature Prioritization, Customer Feedback, Cross-Functional Leadership, Agile Delivery, Stakeholder Management, P&L Ownership, Partnerships, Growth Analytics

ADDITIONAL PROJECTS

Question Answering System / *Python, PyTorch, NumPy, SciPy, NLP*

Academic Project

- Built an NLP-based question-answering system with data preprocessing, model training, and inference workflows for unstructured text.
- Implemented repeatable preprocessing and model-evaluation workflows to support structured experimentation and performance analysis.

Hate Crime Locator / *Kotlin, Firebase, Geolocation, Data Visualization*

Academic Project

- Developed an Android application that combined geospatial data and real-time visualization to help users explore reported incidents by location.
- Integrated Firebase for real-time data storage and location-based updates across mobile sessions.

